# IN A COMPOSITE OF TWO MORPHISMS, IF TWO OF THE THREE ARE UNIVERSAL HOMEOMORPHISMS THEN SO IS THE THIRD

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-permanence-universal-homeomorphism is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 11433-11439 and proof span 11441-11488 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* The proof contains two large xymatrix diagrams of base changes; both are copied verbatim and the retained chapter preamble loads Xy-pic.

*Editorial note.* This card was previously skipped because the shared gate substring-matched my chapter name against the published items of more-morphisms.tex; the gate was fixed to exact file-name equality by the Lead and the candidate was re-gated (RESULT=OK) before being opened.

*Editorial note.* 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- The composite of two universal homeomorphisms is a universal homeomorphism, cited from the same chapter. Cited at proof lines 11441-11442. Disposes of the case in which the two factors are universal homeomorphisms.
- A homeomorphism is an affine morphism, and an affine morphism is separated, both cited from the same chapter. Cited at proof lines 11479-11482. Shows that the auxiliary morphism in the last case is a closed immersion.
- A section of a morphism of schemes is an immersion, cited from the schemes chapter. Cited at proof lines 11483-11485. Turns the identity relation between the auxiliary morphism and the homeomorphism into closedness of the auxiliary morphism.
- The characterization of universal homeomorphisms, used here to conclude from bijectivity plus being a closed immersion. Cited at proof lines 11486-11488. Gives the final conclusion in the last case.

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*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 01S7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2328–2328. Same-chapter reference used in the retained author text at source line(s) 1923, 11481. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0CEV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15590–15590. Same-chapter reference used in the retained author text at source line(s) 11443, 12157. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 04DE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5724–5724. Same-chapter reference used in the retained author text at source line(s) 11372, 11416, 11480, 12247. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 04DF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5725–5725. Same-chapter reference used in the retained author text at source line(s) 11301, 11487, 11529, 11568, 11780, 11886, 14718, 14908. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 01KT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2062–2062. Same-chapter reference used in the retained author text at source line(s) 242, 1012, 2388, 6788, 11217, 11484, 12322, 12920. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $f : X \to Z$ be the composition of two morphisms $g : X \to Y$ and $h : Y \to Z$. If two of the morphisms $\{f, g, h\}$ are universal homeomorphisms, so is the third morphism.*

*Proof.* If both of $g$ and $h$ are universal homeomorphisms, so is $f$ by Lemma 0CEV.

Suppose both of $f$ and $g$ are universal homeomorphisms. We want to show that $h$ is also. Now base change the diagram along an arbitrary morphism $\alpha : Z' \to Z$ of schemes, we get the following diagram with all squares Cartesian:

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & Y' & \xrightarrow{h'} & Z' \\ \downarrow & & \downarrow & & \downarrow \\ X & \xrightarrow{g} & Y & \xrightarrow{h} & Z. \end{array}$$

Our assumption implies that the composition $f' = h' \circ g' : X' \to Z'$ and $g' : X' \to Y'$ are homeomorphisms, therefore so is $h'$. This finishes the proof of $h$ being a universal homeomorphism.

Finally, assume $f$ and $h$ are universal homeomorphisms. We want to show that $g$ is a universal homeomorphism. Let $\beta : Y' \to Y$ be an arbitrary morphism of schemes. We get the following diagram with all squares Cartesian:

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & Y' & & \\ \downarrow & & \downarrow{\scriptstyle \gamma} & & \\ X'' & \xrightarrow{g''} & Y'' & \xrightarrow{h''} & Y' \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle h\circ\beta} \\ X & \xrightarrow{g} & Y & \xrightarrow{h} & Z. \end{array}$$

Here the morphism $\gamma : Y' \to Y''$ is defined by the universal property of fiber products and the two morphisms $id_{Y'} : Y' \to Y'$ and $\beta : Y' \to Y$. We shall prove that $g'$ is a homeomorphism. Since the property of being a homeomorphism has 2-out-of-3 property, we see that $g''$ is a homeomorphism. Staring at the top square, it suffices to prove that $\gamma$ is a universal homeomorphism. Since $h''$ is a homeomorphism, we see that it is an affine morphism by Lemma 04DE and a fortiori separated (Lemma 01S7). Since $h'' \circ \gamma$ is the identity, we see that $\gamma$ is a closed immersion by Schemes, Lemma 01KT. Since $h''$ is bijective, it follows that $\gamma$ is a bijective closed immersion and hence a universal homeomorphism (for example by the characterization in Lemma 04DF) as desired. □